HOMEWORK ASSIGNMENT 8 – Solutions, Revised

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

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QUESTION 1: Rent Comparisons

Are apartments priced differently in San Luis Obispo versus Santa Barbara? To answer this question, we'll take a look at a dataset containing 362 different apartments and their **city** (SLO or SB) and their **rent** (monthly rent in USD), as sourced from *Zillow*[1].

(a) What is one observational unit in the context of this dataset?

Solution: One observational unit is apartment.

(b) Identify the explanatory and response variables, and classify each as either categorical or quantitative.

Solution:

- **Explanatory variable:** city (with levels "SB" and "SLO"). This is a **categorical** variable.
- **Response variable:** rent (in USD/month). This is a **quantitative** variable.

(c) Write the null and alternative hypotheses in symbols and math only, taking care to define all symbols you use.

Solution: Define the following parameters:

- μ_{SLO} : the population mean rent of all apartments in San Luis Obispo
- μ_{SB} : the population mean rent of all apartments in Santa Barbara

We can then write our null and alternative hypotheses as

$$\begin{cases} H_0: & \mu_{SLO} - \mu_{SB} = 0 \\ H_A: & \mu_{SLO} - \mu_{SB} \neq 0 \end{cases}$$

Note: the remainder of this problem will require you to have copy-pasted the data into the Multiple Means Applet. Please follow the instructions outlined in Lecture Activity 16 ("Plate Up!") to accomplish this. The data can be accessed using this link.

(d) Fill in the following table of summary statistics from the dataset.

[1] Data is current as of October 20, 2026, but may contain apartments outside of the stated cities. The data has also not been thoroughly vetted.

San Luis Obispo	Santa Barbara
$n_{SLO} =$ 119	$n_{SB} =$ 243
$\bar{x}_{SLO} =$ 2654.30 USD/month	$\bar{x}_{SB} =$ 4295.23 USD/month
$s_{SLO} =$ 1339.59 USD/month	$s_{SB} =$ 3856.62 USD/month

Table 1: Summary Statistics

(e) Calculate the value of the statistic, using the order (SLO − SB).

Solution: From the applet we get **–1640.928 USD/month**; alternatively, we could calculate

$$\bar{x}_{SLO} - \bar{x}_{SB} = 2654.30 \text{ USD/month} - 4295.23 \text{ USD/month} = \textbf{–1640.93 USD/month}$$

(f) Use at least 10,000 repetitions to obtain an approximate simulation-based *p*-value. **Note:** there may be a slight pause between when you click "Shuffle Responses" and when the applet displays the simulated null distribution, because the dataset is somewhat large.

Solution:

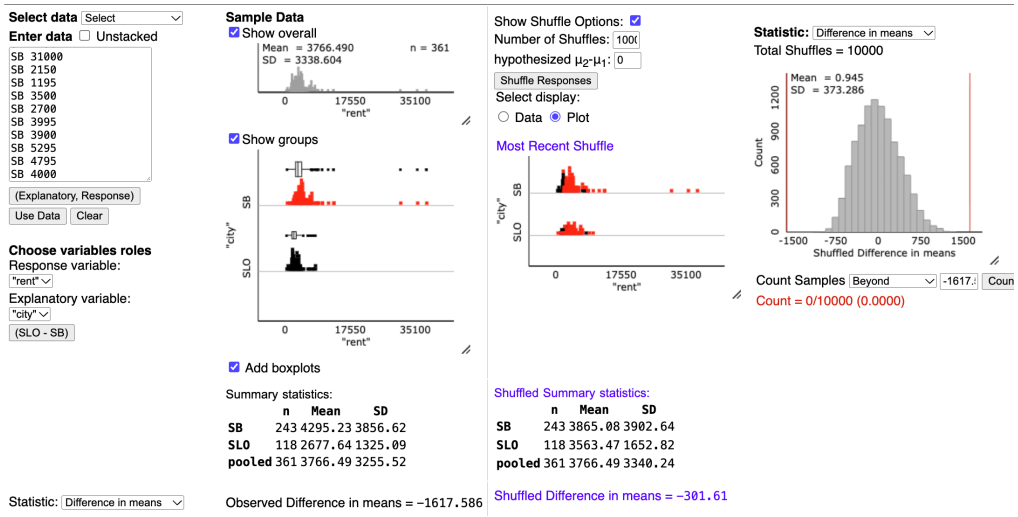


Figure 1: Screenshot from the Multiple Means Applet

My approximate simulation-based *p*-value is 0.00.

(g) Interpret the *p*-value you obtained in Part (f) above.

Solution: The probability of obtaining a difference in sample mean rents (SLO − SB) of –1640.928 US/month or less merely from random assignment is about 0.000, if we assume that the two cities have equal population average apartment rental prices.

(h) Evaluate this *p*-value: How much evidence do the sample data provide against the null hypothesis? Choose one.

- ◯ Not much evidence against the null hypothesis
- ◯ Moderate evidence against the null hypothesis
- ◯ Strong evidence against the null hypothesis
- √ **Very strong evidence against the null hypothesis**

(i) To what population can we generalize the results of this study? Make the best argument you can with the provided data, but also comment on any additional information you think would be pertinent to have.

Solution: ~~It would be nice to have more information about how the sample was collected; in particular, whether or not the 1,794 movies were randomly selected or not. If we assume the sampling procedure was not random, then we have a fairly limited scope of inference - we can really only generalize the results to films similar to those included in the study.~~

Not all apartments get listed on *Zillow*, so we should be wary of generalizing to "all apartments in the two cities." Rather, I would simply say we can generalize our results to the "set of all apartments in SLO and SB that are similar to the ones included in the sample (listed on *Zillow*, similar counts of bedrooms and square footage, etc.)

(j) Given the results of your analysis, provide an overall conclusion in context. Be very clear what you can claim about the population means. IF you have evidence that one population mean is greater/smaller/different than the other, then you need to state which one is greater/smaller/different.

Solution: ~~We have strong/convincing/sufficient evidence to say that the population mean budget of films that PASS Bechdel test is LESS THAN the population mean budget of films that FAIL the Bechdel test, for films similar to the ones included in the study.~~

We have strong/convincing/sufficient evidence to say that the population mean rent of apartments in SLO is DIFFERENT than the population mean rent of apartments in Santa barbara, for apartments similar to the ones included in the study.

If I were grading this problem, here are some things I would have looked for in your answers:

- a clear reference to the parameter - i.e. population means
- use of the context of city and apartment rental price
- a clear statement about which population mean is higher, if either
- an explicit/specific reference to the population (apartments similar to those included in the study)

(k) Explain why we are or are not justified in using a theory-based approach to answer the research question. **Hint:** you may need to go back to the applet and get some additional information about the data!

Solution: Even though we have quite large sample sizes (~~794~~ 243 and 118, both of which are much larger than 20) the **paralell boxplots** from the applet output reveal that the distribution of ~~domestic gross earnings within each category (pass or fail)~~ rents within each category (SLO or SB) are highly right-skewed (as indicated by the long right whiskers, and outliers on the right). Hence, the **valid-**

ity conditions for a two-sample *t*-test are not met, and we are **not** justified in using a theory-based approach to answer the research question.

(l) Critique this study. In particular, comment on how well you think the study is actually addressing the research question.

Solution: This question is intentionally a bit open-ended! Here are some points I thought of:

- Using the population *mean* as a summary metric of a "typical" rent price is probably **not** a good idea - testing the population *medians* is probably a slightly better choice when trying to answer the question "are apartments priced differently in San Luis Obispo versus Santa Barbara?"
- The original research question itself is pretty broad, and could probably do with some tweaking. For example, it might be nice to take into account the *size* of the apartments - we might expect Santa Barbara to have more 3- and 4-bedroom apartments than San Luis Obispo, which we may want to try and factor in.
- Not every apartment gets listed on *Zillow*; for example, many smaller rental companies opt to list their available apartments on their websites as opposed to posting to sites like *Zillow*. As such, our sampling frame is probably much smaller than the target population, leading to a slightly hindered scope of inference.

(m) Return to the applet and check the "95% CI(s) for difference in means" button, and report the resulting confidence interval. Also interpret this interval.

Solution: The reported interval is **(–2160.19, –1074.98)**.

We are 95% confident that the true population average rent among SLO apartments is between 1074.98 and 2160.19 dollars per month LOWER than the true population average rent among SB apartments.

Some things I would have looked for, if I were grading this problem:

- Clearly stating the confidence level (95%)
- Referring to the parameters (population average)
- Using context (monthly rent of apartments)
- Specifying which population has a HIGHER or LOWER population average
- Clearly specifying the population(s)

QUESTION 2: Height and Handedness

Do right-handed and non-right-handed US adults[2] have different heights? To answer this question, we'll use our data from the Welcome Survey. As a reminder: I sent out the welcome survey to my three STAT 1110 sections. After cleaning up the data (removing missing values, correcting mis-inputs, etc.) I was left with 82 responses. We'll consider the two variables to be "**handendess**" (with values "right-handed" or "not right handed") and "**height**" (as measured in inches). Here is the summary of the data:

[2] For this problem, we'll classify anyone college-age or older to be an "adult"

Right-Handed	Not Right-Handed
$n_{\text{Right}} = 67$	$n_{\text{NotRight}} = 15$
$\bar{x}_{\text{Right}} = 65.984$ in	$\bar{x}_{\text{Right}} = 64.850$ in
$s_{\text{Right}} = 2.823$ in	$s_{\text{NotRight}} = 3.652$ in

Table 2: Summary Statistics

Additionally, we'll assume the variable "height" has a symmetric distribution regardless of a person's handedness, which will allow us to proceed with a theory-based approach to inference even though one group has fewer than 20 observations.

Whenever necessary, use the order (Right − NotRight).

(a) Write down the null and alternative hypotheses in symbols and math only.

Solution:

$$\begin{cases} H_0: & \mu_{\text{Right}} - \mu_{\text{NotRight}} = 0 \\ H_A: & \mu_{\text{Right}} - \mu_{\text{NotRight}} \neq 0 \end{cases}$$

(b) Provide a verbal description of any symbols you used in Part (a) above.

Solution:

- μ_{Right}: the population average height of all right-handed US adults
- μ_{NotRight}: the population average height of all non-right-handed US adults

(c) Use the Theory-based Inference applet to carry out a two-sample t-test to answer the research question. To show your work include a screen shot of your output showing the (1) null distribution (curve), (2) standardized statistic and (3) p-value. Save the screen shot as a picture and upload it to this question.

Solution:

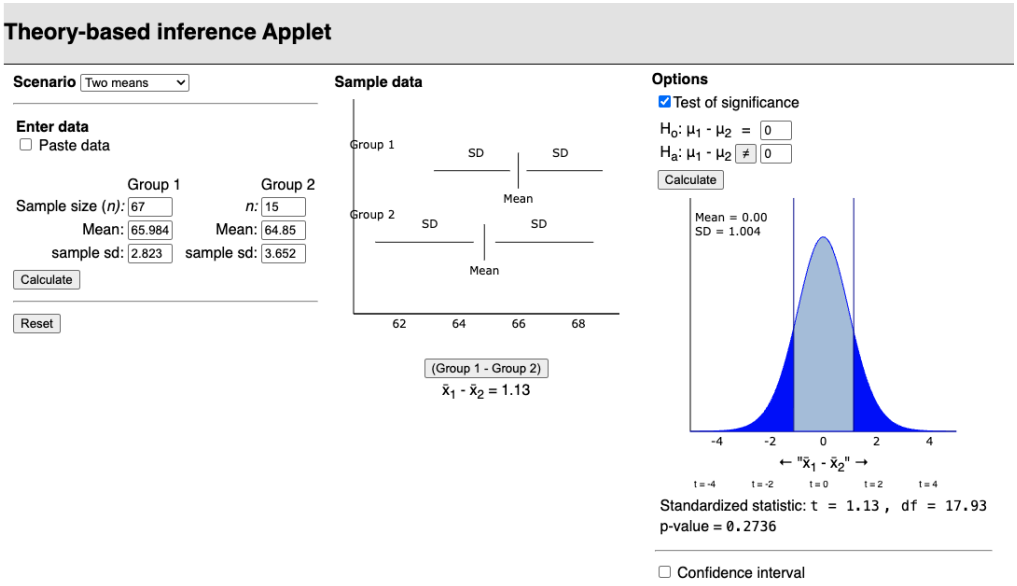


Figure 2: Screenshot from the Theory-Based Inference Applet

The theory-based approximate *p*-value is **0.2736**.

(d) Should we reject or fail to reject the null hypothesis?

◯ Reject the null hypothesis in favor of the alternative hypothesis

√ **Fail to reject the null hypothesis**

(e) Navigate back to the Theory-Based Inference applet, and construct a 95% confidence interval for the difference in population average heights. Interpret your interval.

Solution: (-0.9760, 3.2440).

We are 95% confident that the difference (Right — NotRight) in population average heights between right-handed and non-right-handed US adults is between -0.9760 inches and 3.2440 inches.

Equivalently: we are 95% confident that the population average height of right-handed US adults is between -0.9760 inches and 3.2440 inches higher than the population average height of non-right-handed US adults.

(f) A student has made the following claim:

> Because the sample (STAT 1110 students in Dr. Ethan's sections) was not obtained through random sampling, we are not able to make cause and effect claims.

Do you agree with the student's reasoning? Explain.

Solution: The student has made two correct statements, but linked them together incorrectly. In particular: it is true that the sampling method was not random, and it is also true that we cannot make causal claims in this study. But these two things are not linked! The reason we cannot make causal claims has nothing to do with what sampling method was used - it's because there was no random assignment. Again, remember the table from the end of Lecture Activity 11.